Total_forecast

13M

Actual Units Last 16 Days

13.32M

Busiest Forecast Day

Sunday 20 August 2...

Quietest Forecast Day

Thursday 24 August ...

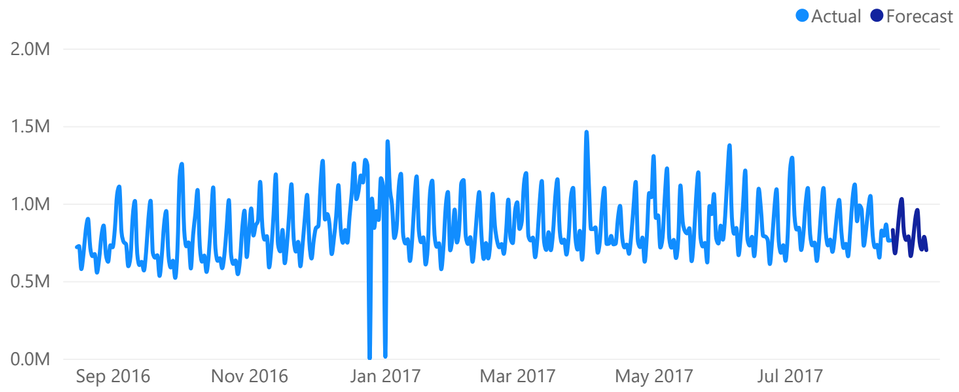
Stores Covered

54

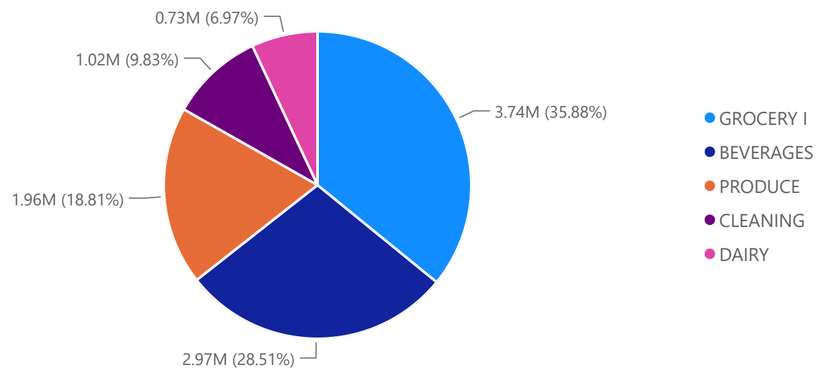
Product Families Covered

33

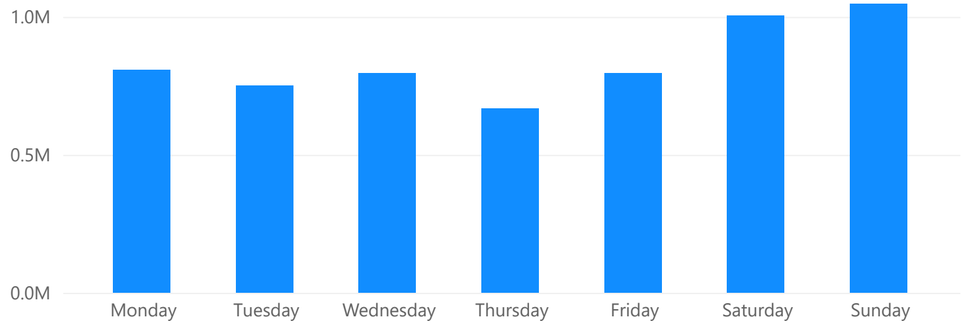
Sales and forecast trend



Share of forecast units among the top 5 families



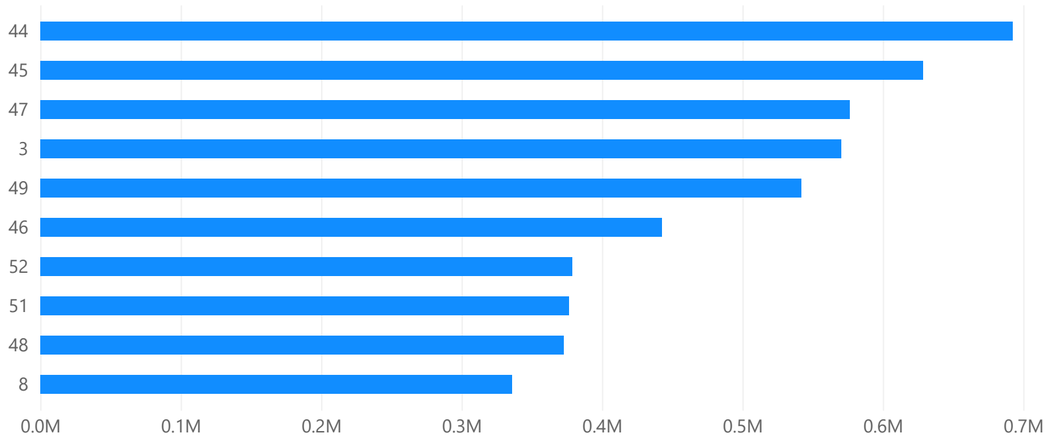
Avg Daily Units by weekday



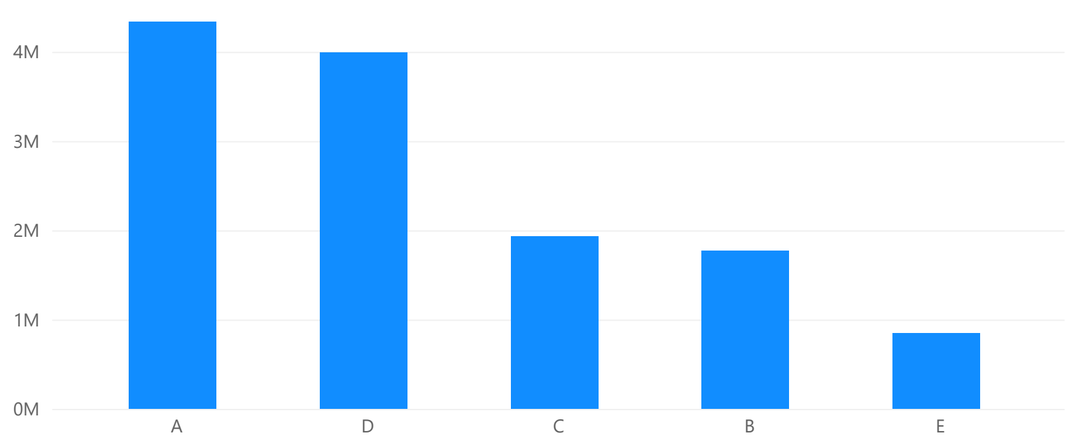
Daily forecast, 16 to 31 August 2017

Year	Quarter	Month	Day	weekday	Sum of forecast_units
2017	Qtr 3	August	16	Wednesday	829,589.74
2017	Qtr 3	August	17	Thursday	678,987.34
2017	Qtr 3	August	18	Friday	797,932.69
2017	Qtr 3	August	19	Saturday	952,283.04
2017	Qtr 3	August	20	Sunday	1,029,990.94
Total					**12,869,361.44**

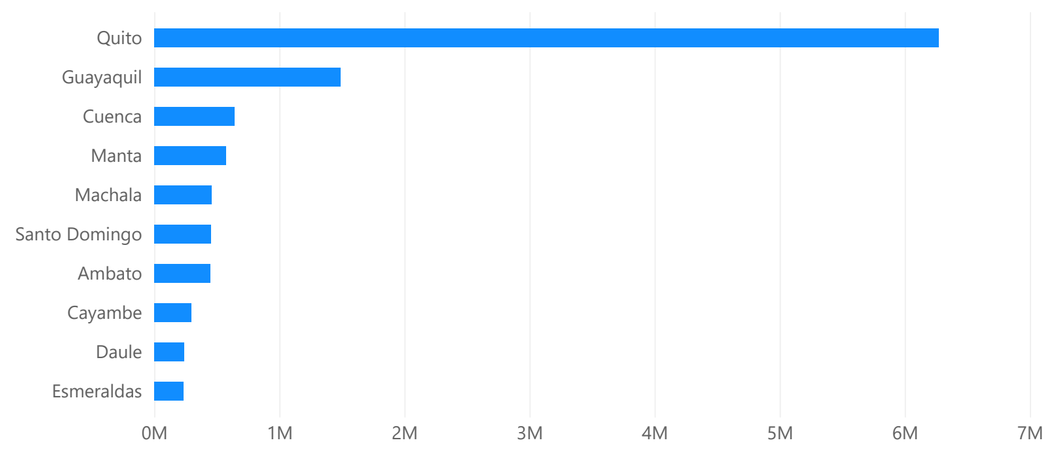
Forecast by store



Sum of forecast_units by store_type



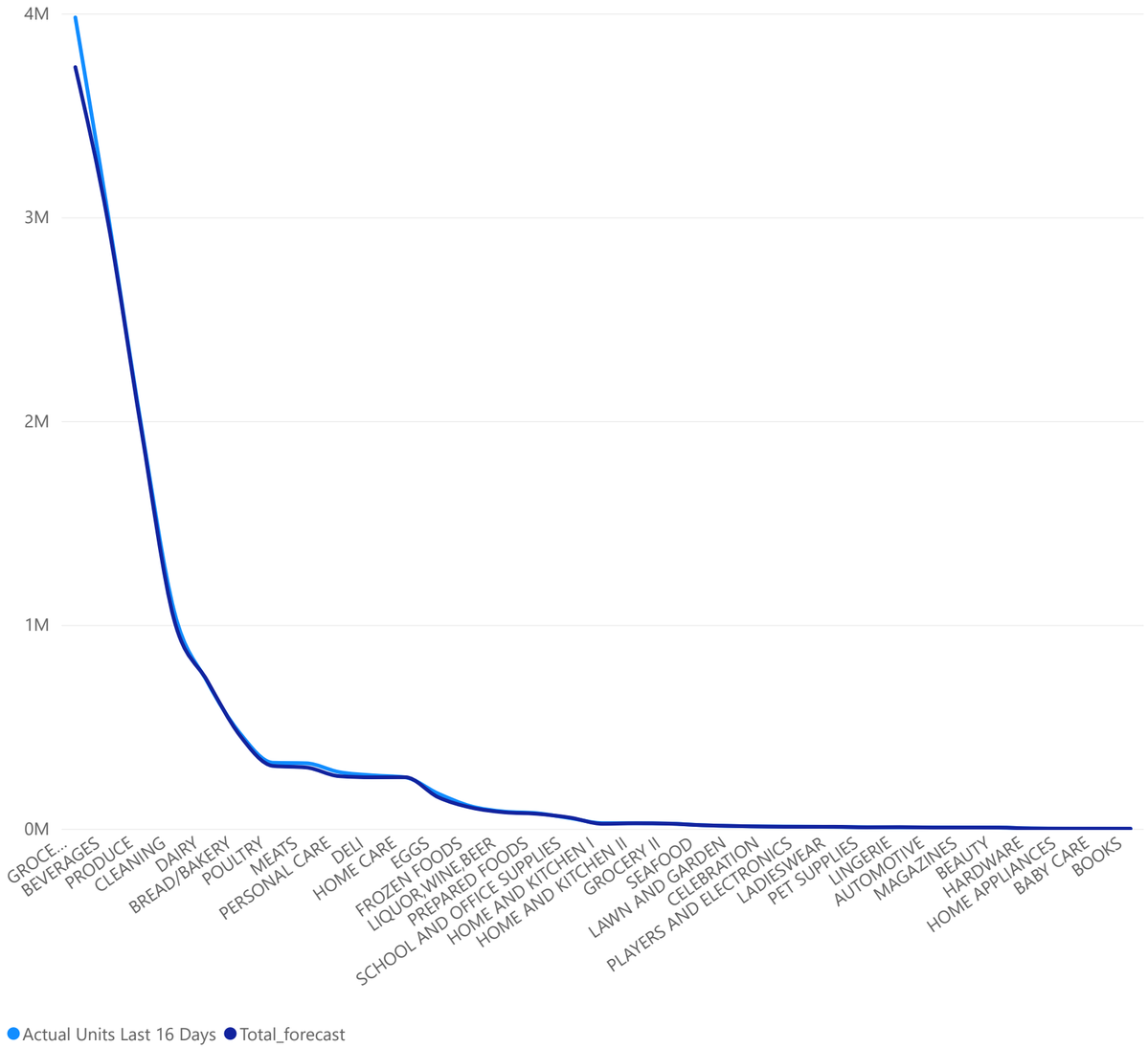
Sum of forecast_units by city



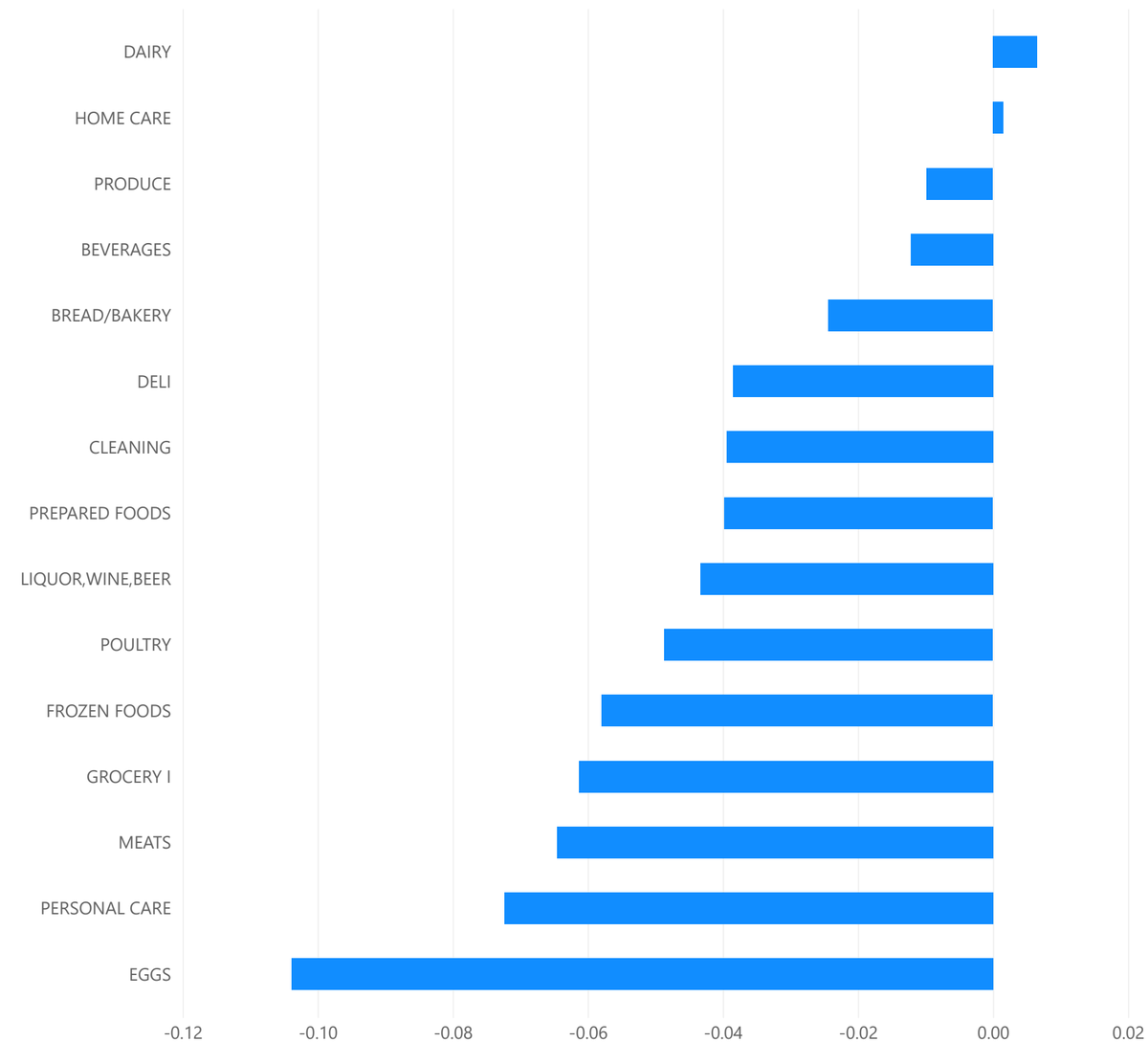
Forecast units: top stores by top families

store_nbr	BEVERAGES	BREAD/BAKERY	CLEANING	DAIRY	GROCERY II	HOME CARE	MEATS	PERSONAL CARE
3	138,402.24	19,923.69	31,635.68	34,128.66	1,288.64	8,555.98	12,385.49	8,708.35
7	65,536.01	11,098.63	16,298.16	21,098.57	602.80	3,993.25	7,628.80	3,974.17
8	75,672.47	13,345.38	19,222.99	22,795.20	616.92	5,674.95	7,314.92	5,224.54
44	160,088.89	26,351.60	40,700.93	40,347.89	1,458.72	11,865.44	14,549.60	10,063.65
45	147,921.45	19,603.70	41,015.17	35,950.54	1,972.31	10,725.98	18,387.13	11,666.13
46	97,156.01	14,553.03	35,457.24	24,124.92	748.31	10,120.46	15,795.85	11,347.00
47	139,962.35	18,361.59	36,750.49	33,972.39	947.68	10,591.22	14,523.81	11,704.62
49	115,794.43	19,560.67	29,360.34	35,936.16	1,117.38	8,112.64	12,931.65	8,354.15
51	77,862.15	14,891.47	23,000.07	26,125.86	653.90	5,445.26	10,340.80	4,825.37
52	78,681.24	14,004.55	26,094.05	19,879.90	951.91	6,184.68	8,015.80	5,852.68
Total	**1,097,077.24**	**171,694.31**	**299,535.12**	**294,360.09**	**10,358.57**	**81,269.86**	**121,873.85**	**81,720.66**

Actual Units Last 16 Days and Total_forecast by family



Change vs Last 16 Days by family



WAPE Ensemble Chain Total

3.87%

WAPE Ensemble Store Family

11.61%

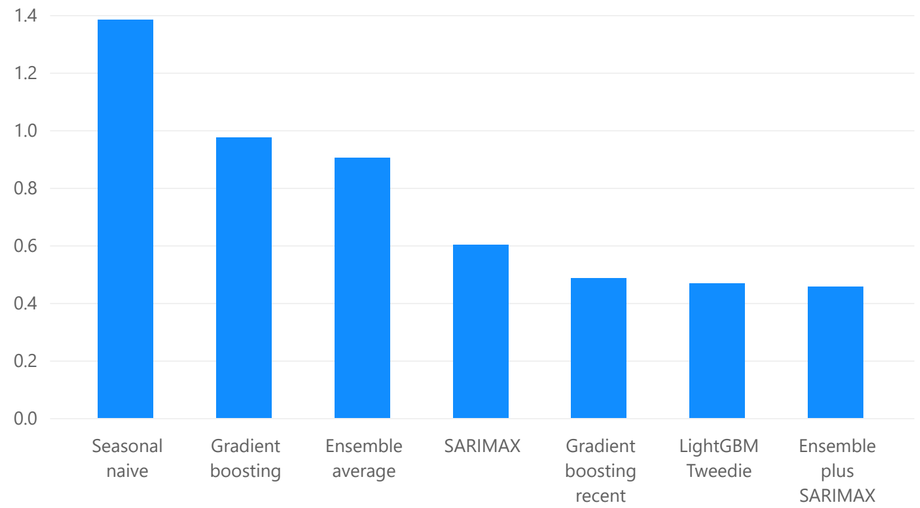
WAPE Naive Store Family

16.96%

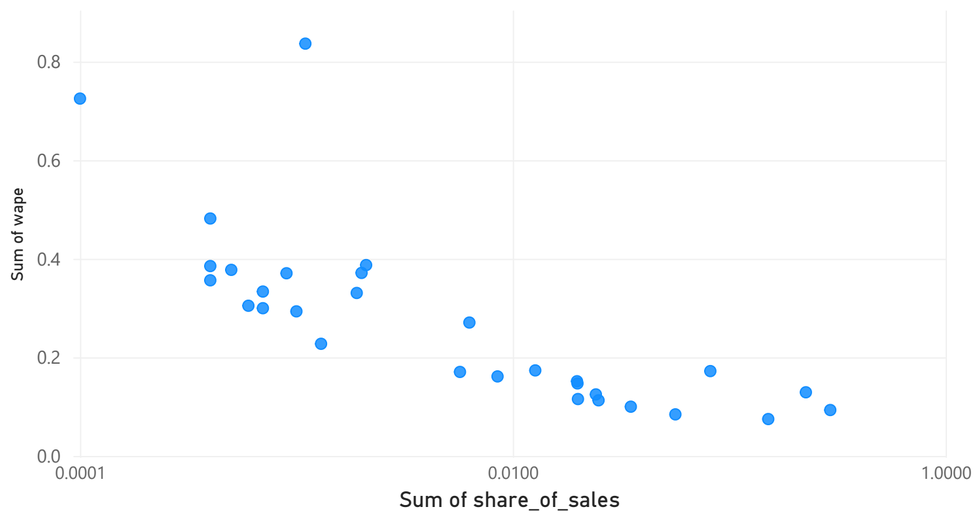
WAPE SARIMAX Chain Total

7.03%

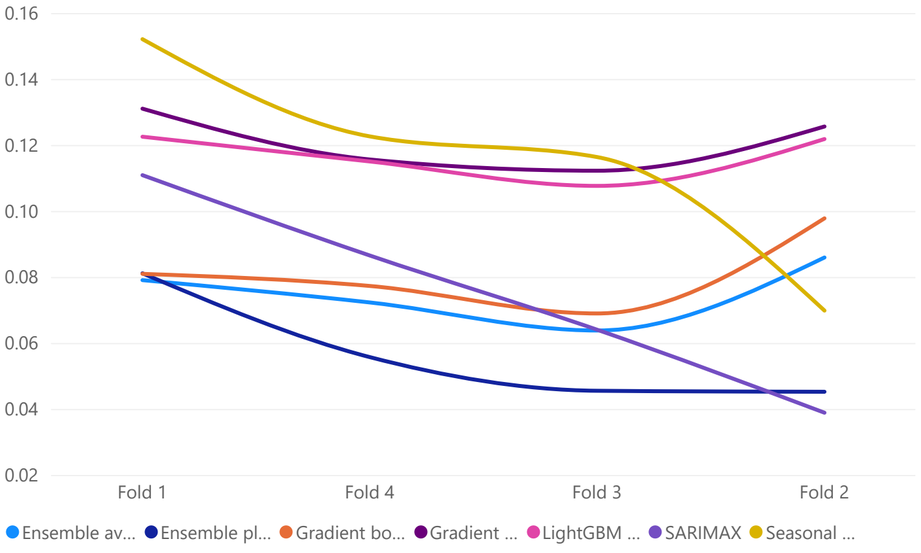
Error by model at each level, lower is better.



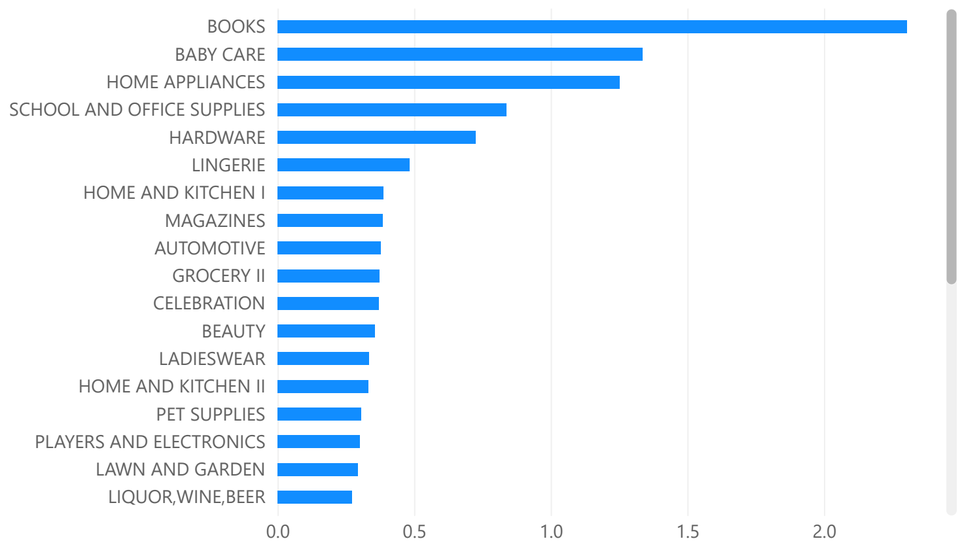
Large families are forecast well, the biggest errors sit in tiny ones.



Average of WAPE by fold and model



Error by store type.



Store sales forecasting for Corporacion Favorita

What this report shows: A 16 day forecast of units sold for 54 stores and 33 product families, covering 16 to 31 August 2017, with the last 365 days of real sales for comparison.

Data: Six retail files: sales, stores, oil prices, holidays and events, transactions and promotions. The unit is items sold, not money.

Method: An average of three gradient boosting models, built only from sales history at least 16 days old and checked on four time ordered folds of 16 days. SARIMAX was tested as a comparison.

Accuracy: About 11.6 percent error for one store and product on one day, and about 3.9 percent for the chain daily total. WAPE is total absolute error divided by total actual sales.

limits: Units, not revenue. A 16 day horizon. Shocks such as a new competitor or supply problems are not captured.